

QA AI Platform

AI-powered Requirement Analysis and Test Case Generation Platform



Quick Start

Prerequisites

- Python 3.11+
- Telegram Bot Token
- DeepSeek API Key or Local LLM Gateway

Installation

```
git clone <repository>

cd qa-ai-platform

python -m venv .venv

.venv\Scripts\activate

pip install -r requirements.txt
```

Configuration

Create `.env`

```
LLM_PROVIDER=DEEPSEEK

DEEPSEEK_API_KEY=xxxxxx

DEEPSEEK_MODEL=deepseek-v4-flash

TELEGRAM_BOT_TOKEN=xxxxxx
```

Run Telegram Bot

```
python -m bot.telegram_bot
```

Verify LLM Connection

```
python test_llm.py
```



Available Commands

Requirement Creation

```
/generate_text
```

Create requirement from text.

```
/generate <ticket_id>
```

Create requirement from Jira ticket.

Requirement Analysis

```
/analyze <ticket_id>
```

Run:

- Requirement Analysis
 - Clarification Generation
 - Requirement Summary
-

Requirement Management

```
/requirements
```

List requirements.

```
/status <ticket_id>
```

Show status.

```
/rename <ticket_id>
```

Rename requirement.

```
/delete <ticket_id>
```

Delete requirement.

```
/delete_all
```

Delete all requirements.

Requirement Refinement

```
/add_text <ticket_id>
```

Add additional requirement notes.

Reporting

```
/report
```

Show:

- AI usage
- Token usage

- Processing time
- Generated assets

Business Overview

QA AI Platform helps QA teams:

- Analyze requirements
- Identify requirement gaps
- Generate clarification questions
- Build test coverage
- Generate test cases
- Maintain traceability
- Improve test quality

Goal:

```
graph TD;
  Requirement --> Coverage;
  Coverage --> QualityTestCases[Quality Test Cases];
```

Core Features

Requirement Intelligence

- Requirement Analysis
- Requirement Items
- Clarifications
- Requirement Summary

Test Design

- Test Scope Generation
- Scenario Generation
- Test Case Generation

Quality Review

- Coverage Review

- Test Case Improvement
- Final Coverage Review

Traceability

```
Requirement
  ↓
Scenario
  ↓
Test Case
```

Export

- Excel Export
- Coverage Matrix

System Architecture

Main Components

```
Telegram Bot
  ↓
FastAPI
  ↓
LangGraph Workflow
  ↓
LLM Provider
  ↓
Requirement Artifacts
```

See:

- Architecture Diagram.md
- Requirement Lifecycle State Machine.md

Workflow

```
Requirement
  ↓
```

```
Requirement Analysis
  ↓
Clarification Questions
  ↓
Requirement Summary
  ↓
Test Scope
  ↓
Scenario Generation
  ↓
Test Case Generation
  ↓
Coverage Review
  ↓
Improve Test Cases
  ↓
Final Review
  ↓
Excel Export
```

Project Structure

```
qa-ai-platform
├── app
│   ├── services
│   ├── utils
│   └── prompts
├── graph
│   ├── nodes
│   └── workflows
├── bot
├── requirements
├── exports
└── tests
```

Supported LLM Providers

DeepSeek

```
LLM_PROVIDER=DEEPSEEK  
DEEPSEEK_MODEL=deepseek-v4-flash
```

Local Enterprise Gateway

```
LLM_PROVIDER=LOCAL  
  
LOCAL_LLM_URL=http://localhost:xxxx/v1/chat/completions  
  
LOCAL_LLM_MODEL=claude-sonnet-4.6
```

Metrics & Reporting

Tracked metrics:

- Requirements generated
- Scenarios generated
- Test cases generated
- Improvement cycles
- AI requests
- Model usage
- Token consumption
- Processing duration

Roadmap

MVP1 (Current)

- ✓ Requirement Analysis
- ✓ Clarifications
- ✓ Requirement Summary

- ✓ Test Scope
 - ✓ Scenario Generation
 - ✓ Test Case Generation
 - ✓ Coverage Review
 - ✓ Improve Test Cases
 - ✓ Traceability Matrix
 - ✓ Excel Export
 - ✓ Telegram Integration
-

MVP2

- Enhanced Jira Integration
 - Requirement Change Impact Analysis
 - AI Test Data Generation
 - Teams Integration
 - Knowledge Base
-

MVP3

- Auto Jira Synchronization
 - Risk-based Testing
 - Regression Impact Analysis
 - Multi-Agent QA Copilot
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Documentation

Additional documents:

- README.md
 - Architecture Diagram.md
 - Requirement Lifecycle State Machine.md
 - Security Policy.md
 - MVP Roadmap.md
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License

Internal project.